

PACKET SCRAMBLER

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Date: 19.08.2026

Program Description

This program performs four stages:

1. Validate the packet.
2. Scramble the packet using slicing.
3. Insert SYNC-BIT and remove zeros.
4. Check memory integrity using unpacking.

Question 1 – Input Validation and Test Data

```
1 # question 1 -- input validation and test data
2
3 packet = [2,6,9,8,0,5,45,0,90,23]
4
5 if packet and len(packet) >= 10:
6     print("Validation passed: Processing packet ....")
7 else:
8     print("Validation failed : Packet is empty or too short.")
9
10 print("INITIAL PACKET", packet)
```

Question 2 – The Middle-Out Swap

```
1 # question 2 -- the middle-out swap
2
3 midpoint = len(packet)//2
4 front_half = packet[:midpoint]
5 back_half = packet[midpoint:]
6
7 scrambled = back_half[::-1] + front_half
```

```

8
9 print(id(packet) == id(front_half))
10
11 print("AFTER STAGE 2 SCRAMBLED", scrambled)

```

Question 3 – In-Place Correction

```

1 # question 3 -- in place correction
2
3 middle_index = len(scrambled) // 2
4
5 if type(scrambled[middle_index]) is int:
6     scrambled.insert(middle_index+1, "SYNC-BIT")
7
8 print("AFTER SYNC BIT INSERTION ", scrambled)
9
10 while 0 in scrambled:
11     scrambled.remove(0)
12
13 print("FINAL SCRAMBLED AFTER ZERO REMOVAL", scrambled)

```

Question 4 – Memory Integrity Check

```

1 # question 4 -- memory integrity check
2
3 first, *middle, last = scrambled
4
5 print(
6     f"HEADER: {first}  "
7     f"FOOTER: {last}  "
8     f"BODY LENGTH: {len(middle)}"
9 )

```

Complete Program

```

1 ''' packet_scrambler
2 Author : Sneha Gautam
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```

```

4
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6
7 1. Validate the packet
8 2. Scramble the packet using slicing
9 3. Insert SYNC-BIT and remove zeros
10 4. Check memory integrity using unpacking
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60     f"BODY LENGTH: {len(middle)}"
61 )

```